

Arinze Elenasulu

Python Backend Engineer | AI & Automation

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Professional Summary

I have been writing Python since 2020 and it is my strongest language. I build backend systems that people actually rely on every day: APIs, databases, automation workflows, and AI tooling. I care about clean code, honest testing, and shipping software that does not need a prayer circle to stay up. My background sits where backend engineering meets practical AI: I have built production SaaS features, trained and fine-tuned models, and designed agent infrastructure that runs 24/7 on a cloud VPS. I am looking for a role where I can build systems that matter, learn fast, and be a genuinely good person to have on the team.

Core Skills

- Backend Engineering:** Python (primary, since 2020), FastAPI, SQLAlchemy, Alembic, PostgreSQL, REST API design, async/await, JWT authentication, role-based access control, schema design, query optimisation
- AI and Automation:** HuggingFace, PyTorch, RAGAS, model fine-tuning, LLM evaluation, self-managed AI agent infrastructure, browser automation, agent orchestration, MCP servers
- Testing and Quality:** Advanced pytest (fixtures, parametrisation, integration coverage), structured test design for backend and AI workflows, test-driven development, API test automation
- Infrastructure and DevOps:** Docker (multi-stage builds, docker-compose), Linux, Git/GitHub, CI/CD, structured logging, input validation
- Additional Languages:** Java (Spring Framework, EJB), Flutter/Dart, C++, C (from formal training; Python is my strongest)

Education

- BSc Information Technology**, Middlesex University London (June 2026)
- Advanced Diploma in Software Engineering (ADSE Java)**, Aptech Computer Education, Lagos (September 2022 to August 2024)
A structured two-year programme combining classroom training with hands-on project work across four semesters. Covered Linux, Python, C++, C, Java, Spring Framework, EJB, Flutter/Dart, MERN and MEAN stacks. Python became my primary language during this period.

Experience

Software Engineer, Vigilis

August 2024 to February 2026 · Lagos, Nigeria

- Built and maintained the backend for a pharmacy operations platform handling sales, inventory, payments, and reporting across multiple locations. The system was used day to day by pharmacy staff.
- Designed the full data model and API layer (FastAPI, PostgreSQL, SQLAlchemy) with role-based access, JWT auth, and Alembic migrations that kept the database evolving safely as features shipped.
 - Sat with pharmacy staff directly to understand their manual processes. Turned paper workflows and mental checklists into software that cut daily admin time and reduced stock discrepancies.
 - Built the invoice lifecycle: create, add line items, finalise with automatic stock deduction, cancel with stock reversal. Every state transition validated and audited.

- Added structured logging, input validation, and business rule enforcement that caught errors at the API boundary before they could corrupt data downstream.
- Owned the backend end to end: feature planning, database design, deployment, monitoring, and production bug fixes. No handoff.

AI and Backend Engineer (Contract), Bucca Hut

February 2025 to May 2025

Built a data analysis and data mining tool for a food business, connecting LLM outputs and mined data insights to real operational decisions. Designed the backend API bridging AI processing with existing workflows. Worked as a contract(extra hands) alongside my position at vigilis Vigilis, delivering the tool from concept to working prototype.

Software Engineer Intern, Aptech Computer Education

September 2022 to August 2024 · Lagos, Nigeria

Completed the Advanced Diploma in Software Engineering (ADSE Java track), a structured two-year programme combining classroom training with hands-on project work.

- Covered the full software engineering stack across four semesters: Linux system administration, Python, C++, C, Java, Spring Framework, Enterprise Java Beans (EJB), Flutter/Dart for mobile, and both MERN and MEAN stacks for full-stack web development.
- Built multiple projects across different technology stacks as part of the programme curriculum, gaining practical exposure to how different languages and frameworks solve similar problems.
- Developed strong fundamentals in object-oriented programming, database design, and software architecture that carried into every role since.
- Python became my primary language during this period and has remained so. I have been studying and building with it continuously since 2020.

Selected Projects

Pharmax, Pharmacy Management Platform

Python, FastAPI, PostgreSQL, SQLAlchemy, Alembic

Built a SaaS for an AI-integrated pharmacy operations platform handling product inventory, sales, invoicing, and reporting. The backend skeleton is public on GitHub; the full SaaS code is proprietary.

- Products CRUD with multi-unit pricing (tablets, packs, strips).
- Full invoice lifecycle: draft to finalise (auto stock deduction) to cancel (stock reversal).
- Stock adjustments with audit trail and reorder-level tracking.
- Role-based access with JWT authentication.
- Comprehensive pytest suite covering endpoints, business logic, and edge cases. Used RAGAS for LLM evaluation workflows.

FRAME, Typed Project Memory Format for AI Agents

Python, Markdown/YAML, Multi-Agent

github.com/haxsysgit/FRAME

Designed a portable standard that gives AI coding agents structured project context across sessions. Five-part memory model (Facts, Rules, Acts, Map, Expect) capturing project truth, agent rules, session history, module relationships, and expected outcomes. Reduces repeated prompting, context rot, and full-repo rescans across agent sessions. FRAME is the contract; implementations earn alignment with it.

Haxaml, AI Agent Governance and Project Memory

Python, MCP, FRAME, System design

github.com/haxsysgit/haxaml · pypi.org/project/haxaml

Built and published a governance runtime that enforces project rules for AI coding agents. Five-stage pipeline from Admission to Verification to Recording. MCP server so AI coding tools can read and write FRAME files during a session. Published on PyPI with documentation and setup guides.

CaseFRAME, Financial Crime Investigation Continuity

Python, FRAME model

github.com/haxsysgit/CaseFRAME

Applied the FRAME memory model to financial-crime investigation continuity, giving investigators structured context persistence across long-running cases.

HaxJobs, Automated Job Search Pipeline

Python, FastAPI, SQLite, React, TypeScript

github.com/haxsysgit/Haxjobs

Built an end-to-end automated job discovery and application platform. Multi-platform discovery across Lever, Ashby, and Greenhouse APIs. Pipeline evaluates job fit against my profile using a deterministic classifier and produces structured application packs. Web dashboard for pipeline monitoring with real-time job status views. Runs on self-managed 24/7 infrastructure on a cloud VPS.

Additional Information

- ▶ Work authorization: currently on a UK student visa, graduating June 2026. Planning to apply for a Graduate visa (2-year work right). No immediate sponsorship required.
- ▶ Availability: available to start immediately.
- ▶ Location preference: London, Manchester, Leeds, Remote UK, Hybrid UK.